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PERSONALIZED CONNECTIVITY SERVICE OFFERINGS

Abstract

Methods, systems, and devices are described for providing selectable service offerings to mobile users via mobile terminals over a satellite communication system. In embodiments, a database of access profiles for accessing a network via the wireless communication system may be maintained, wherein each access profile corresponds to a service offering of a plurality of service offerings selectable via the plurality of devices and is associated with a set of network destinations within the network. In embodiments, a request is received from a device of the plurality of devices for a particular service offering of the plurality of service offerings. An access profile may be identified from the database that is associated with the particular service offering. Communications may be permitted between the device and the respective set of network destinations associated with the identified access profile.

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Background/Summary

CROSS REFERENCE [0001] The present Application for Patent is a continuation of U.S. patent application Ser. No. 18/245,447 by O'Sullivan et al., entitled “Personalized Connectivity Service Offerings” filed Mar. 15, 2023, which is a 371 national phase filing for International Patent Application No. PCT/US2020/051131 by O'Sullivan et al., entitled “Personalized Connectivity Service Offerings” filed Sep. 16, 2020, each of which is assigned to the assignee hereof, and each of which is expressly incorporated by reference in its entirety herein.

BACKGROUND
[0002] In a wireless communications system, a satellite communication system may provide network connectivity for a number of devices communicatively coupled with a satellite terminal of the wireless communication systems. The network connectivity offered to the devices may be binary (e.g., all or nothing), such as being connected to all of a network such as the Internet or being disconnected. Services which are offered to the devices within a single carrier are typically the same regardless of any qualities or characteristics of the devices. Providing complete network access to a device may be costly and an inefficient use of available radio frequency (RF) spectrum resources in the satellite communications system. Furthermore, not every service offered may be applicable to every device.
SUMMARY
[0003] The described features generally relate to providing personalized service offerings for devices connected to a satellite terminal in a satellite communications system. In particular, connectivity solutions are provided to each customer or passenger associated with a device which are curated particularly for that customer or passenger. The techniques described herein provide a la carte, passenger-customizable products, services, and applications at their own devices while onboard a mobile platform. When a service is requested by the device, a micro-session may be established for the service between the device and a network providing the service via the satellite communication system. The micro-session may restrict network traffic to a set of network destinations associated with the requested service. This technique enables traffic shaping, reduces the amount of traffic over the satellite communication system, reduces cost, and provides the passenger with a more customized experience.
[0004] Various content and service providers on one or more networks may provide information related to the service offerings to a carrier associated with the mobile platform. Each service offering may be personalized using one or more access profiles. The access profiles may define a set of rules for the plurality of service offerings, including the particular service offering, offered at the plurality of devices within the mobile platform. A service offering manager may be located within the satellite communication system, for example onboard the mobile platform. The service offering manager may compare a requested service offering and a requesting device to the relevant access profile. If there is a match, the satellite communication system is authorized to allow access to the service offering at the device. The satellite communication system may facilitate communications between the network and the device at the particular set of network destinations, to provide the requested service at the device.
[0005] The foregoing has outlined rather broadly the features and technical advantages of examples according to the disclosure in order that the detailed description that follows may be better understood. Additional features and advantages will be described hereinafter. The conception and

specific examples disclosed may be readily utilized as a basis for modifying or designing other structures for carrying out the same purposes of the present disclosure. Such equivalent constructions do not depart from the scope of the appended claims. Characteristics of the concepts disclosed herein, both their organization and method of operation, together with associated advantages will be better understood from the following description when considered in connection with the accompanying figures. Each of the figures is provided for the purpose of illustration and description only, and not as a definition of the limits of the claims.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0006] A further understanding of the nature and advantages of the present disclosure may be realized by reference to the following drawings. In the appended figures, similar components or features may have the same reference label. Further, various components of the same type may be distinguished by following the reference label by a dash and a second label that distinguishes among the similar components. If only the first reference label is used in the specification, the description is applicable to any one of the similar components having the same first reference label irrespective of the second reference label.

[0007] FIG. **1** shows a diagram of a wireless communication system, in accordance with aspects of the present disclosure;

[0008] FIG. **2** shows a diagram of an in-transport media delivery system, in accordance with various aspects of the present disclosure;

[0009] FIG. **3** shows a process flow diagram of an example for providing service offerings, in accordance with aspects of the present disclosure;

[0010] FIG. **4** shows an example screenshot of a dashboard for a plurality of service offerings, in accordance with aspects of the present disclosure;

[0011] FIGS. **5** through **8** show flowcharts illustrating exemplary methods for personalized connectivity service offerings, in accordance with aspects of the present disclosure;

[0012] FIG. **9** shows a block diagram illustrating a satellite communications environment, in accordance with aspects of the present disclosure; and

[0013] FIG. **10** shows a block diagram of a service offering manager, in accordance with aspects of the present disclosure.

DETAILED DESCRIPTION

[0014] A satellite terminal (e.g., a multi-user access terminal) within a satellite communication system may employ a communications antenna to establish a communications link between the satellite terminal and a communications satellite. The communications link may be configured for bi-directional communications (e.g., transmitting and receiving, etc.), or uni-directional communications (e.g., receiving), in some cases. Such a communication link may provide communication services between a ground-based network and a mobile transport that includes a satellite terminal. These communication links may be used to provide services to devices (e.g., user devices such as mobile phones and laptops) for passengers located on the mobile transport.

[0015] A procedure may be used to establish a particular service at a device located onboard a mobile transport that has connectivity via the satellite communication system. The satellite communication system relays network traffic for the particular service between the device and a network. According to aspects of the present disclosure, one or more networks may provide information related to service offerings to a satellite terminal, the service offerings including one or more services of a network. The satellite terminal may provide these service offerings to devices onboard a mobile transport associated with the satellite terminal. The satellite terminal may generate a portal for each device that is customized for that device based on a set of access profiles associated with the service offerings. The portal may show service offerings that are authorized for the device, which may be based on a characteristic of the device or an associated user of the device. Service offerings may differ between customers based on content, price, type, and the like.

[0016] When a customer selects a service of the service offering to use (e.g., using the generated portal), the satellite terminal may establish communications between the device and the relevant network and network destinations for that particular service. That is, acceptance of a service offering whitelists the device to communicate with one or more network destinations associated with the service offering. The satellite terminal may facilitate the service by acting as a relay for the network activity.

[0017] This description provides examples, and is not intended to limit the scope, applicability or configuration of embodiments of the principles described herein. Rather, the following description will provide those skilled in the art with an enabling description for implementing embodiments of the principles described herein. Various changes may be made in the function and arrangement of elements.

[0018] Thus, various embodiments may omit, substitute, or add various procedures or components as appropriate. For instance, it should be appreciated that the methods may be performed in an order different than that described, and that various steps may be added, omitted or combined. Also, aspects and elements described with respect to certain embodiments may be combined in various other embodiments. It should also be appreciated that the following systems, methods, devices, and software may individually or collectively be components of a larger system, wherein other procedures may take precedence over or otherwise modify their application.

[0019] FIG. **1** shows a diagram of a wireless communications system **100**, in accordance with aspects of the present disclosure. Wireless communications system **100** includes a satellite communications environment, which includes a satellite communication system **157** (including one or more communications satellites **121**), a satellite terminal **150**, one or more devices **160**, one or more satellite terminals **136**, a gateway **130**, and one or more networks **140**. Satellite terminal **150** may be multi-user access terminals providing network access connectivity to multiple devices **160** on a mobile platform **170**. Located onboard mobile platform **170** is a service offering manager **172** that provides connectivity to a selected service from a service offering at the devices **160**. The wireless communications system **100** may be connectable to at least one user device **160** and to one or more networks **140** directly, or via one or more network devices **141**.

[0020] One or more communications satellites **121** in satellite communications system **157** may include any suitable type of communication satellite configured for wireless communication with gateway **130** and one or more satellite terminals **150**. In some examples, some or all of communications satellites **121** may be in geostationary orbits, such that their positions with respect to terrestrial devices may be relatively fixed, or fixed within an operational tolerance or other orbital window. In other examples, any appropriate orbit (e.g., low Earth orbit (LEO), medium Earth orbit (MEO), etc.) for one or more satellite(s) **121** of the satellite communications system may be used.

[0021] Satellite terminal **150** may include a satellite terminal communications antenna configured for receiving and transmitting signals **155** from communications satellite **121**. Satellite terminal **150** may be configured for uni-directional or bi-directional communications with one or more communications satellites **121** of satellite communications system **157**.

[0022] Communications satellite **121** may communicate via signals **155** directed towards a service beam coverage area **126** that includes satellite terminal **150**. Service beam coverage area **126** may cover any suitable service area (e.g., circular, elliptical, hexagonal, local, regional, national, etc.) and provide service to any number of satellite terminals **150** located within service beam coverage area **126**. Likewise, communications satellite **121** may communicate via signals **135** directed towards a service beam coverage area **136** that includes one or more satellite terminals **110**. Communications satellite **121** may communicate with gateway **130** by sending and/or receiving signals **135**. Signals **135** may, for example, carry communications traffic for one or more satellite terminals **150** (e.g., relayed by communications satellite **121**), or other communications between the communications satellite **121** and gateway **130**. In some examples, communications satellite **121** may be a multi-beam satellite and may have multiple service beams covering multiple service beam coverage areas, including service beam coverage areas **126** and **136**, which may or may not overlap with adjacent service beam coverage areas.

[0023] Wireless communications system **100** (including satellite communications system **157**) may operate in one or more frequency bands. For example, wireless communications system **100** may operate in the International Telecommunications Union (ITU) Ku, K, or Ka-bands, C-band, X-band, S-band, L-band, and the like. Additionally, satellite terminal **150** may be used in other applications besides ground-based stationary systems, including mobile applications such as boats, aircraft, ground-based vehicles, and the like.

[0024] Satellite terminal **150** may be located on mobile platform **170**, which may be any device, apparatus, or object capable of supporting satellite terminal **150** and of changing location. For example, the mobile platform may be a mobile transport carrier such as an aircraft, a space shuttle, a ship, a vehicle, or the like.

[0025] Gateway **130** may send and receive signals **135** to and from communications satellites **121** of satellite communications system **157** using one or more gateway satellite terminals **110**. A gateway antenna terminal **110** may be two-way capable and designed with adequate transmit power and receive sensitivity to communicate reliably with at least one communications satellite **121**. Gateway **130** may also communicate with one or more networks **140**. One or more networks **140** may be part of or outside of the wireless communication system. The wireless communication system may access the one or more networks **140** using a database of access profiles **145**. The one or more networks **140** may include a local area network (LAN), metropolitan area network (MAN), wide area network (WAN), or any other suitable public or private network and may be connected to other communications networks such as the Internet, telephony networks (e.g., Public Switched Telephone Network (PSTN), etc.), and/or the like. One or more network devices **141** may be coupled with gateway **130** through the one or more networks **140** and may control aspects of wireless communications system **100** in some examples. In various examples, a network device **141** may be co-located or otherwise located nearby gateway **130**, or may be a remote installation that communicates with gateway **130** and/or the one or more network(s) **140** via one or more wired and/or wireless communications links. In some examples, at least some of network devices **141** are servers. Network **140** may include one or more service offerings **167**.

[0026] Aircraft communication equipment **174** may include satellite terminal **150** (which may be, be part of, or include aspects of a multi-user access terminal), a service offering manager **172**, a network access unit **164**, a wireless access point **166**, and user devices **160-*a*** through **160-*d*** (collectively referred to herein as device **160**). The satellite terminal **150** may include one or more mobile terminal antennas, a transceiver **176**, and a modem **178**. All or some of the aircraft communication equipment **162** may be located in an interior of the aircraft.

[0027] Mobile platform **170** includes aircraft communication equipment **174** that includes satellite terminal **150**, including an antenna, which may be an antenna array. Mobile platform **170** may use an antenna of satellite terminal **150** to communicate with satellite communication system **157** via one or more signals **155**. Portions of satellite terminal **150** (e.g., the antenna, transceiver **176**) may be mounted on the outside of the fuselage or other location on the exterior of mobile platform **170**, and may be installed under a radome. In other examples, other types of housings may be used to house portions of satellite terminal **150**. Other portions of satellite terminal **150** may be on the interior of the aircraft (e.g., modem **178**). Satellite terminal **150** may operate in the ITU Ku, K, or Ka-bands. Alternatively, satellite terminal **150** may operate in other frequency bands such as C-band, X-band, S-band, L-band, and the like.

[0028] Data sent over the downlink and uplink to satellite communication system **157** over the one or more signals **155** may be formatted using a modulation and coding scheme (MCS) that may be custom to the satellite or similar to others in the industry. For example, the MCS may include multiple code-points that each are associated with a modulation technique (e.g., BPSK, QPSK, 16 QAM, 64 QAM, 256 QAM, etc.) and a coding rate that is based on a ratio of the coded information bits to a total number of coded bits including redundant information.

[0029] Service offering manager **172** may be located on-board mobile platform **170** and may include a processor **180**, a network interface **182**, and a memory **186**. Service offering manager **172** may manager different service offerings among various devices, such as providing requested service offerings, approving requested service offerings, and whitelisting devices to communicate with associated network destinations when the service offerings are accepted. Processor **180** may execute instructions stored on memory **186** to perform the functions of service offering manager **172**. Memory **186** may store the instructions for the operation of service offering manager **172**, local copies of media content, and an access profile database **145-**b, and one or more device-specific profiles **149**. In some examples, memory **186** stores a database of device specific profiles. Memory **186** may also store an index that may organize or otherwise identify content stored at memory **186**.

[0030] Database of access profiles **145** may include a set of access profiles that define a set of rules for a plurality of service offerings. Each service offering may be authorized for a particular device **160** based on an access profile of a device **160**. The access profile may be based on one or more characteristics of the device **160**. For example, the access profile may show that one or more service offerings may be authorized for the device **160** based on a characteristic of device **160**, an identity of a user associated with device **160**, a subscription of device **160**, a service provider of device **160**, one or more characteristics of mobile platform **170** carrying device **160**, specific to a flight or other travel route, communication information, a reservation identifier, a ticket identifier, an individual membership associated with device **160**, a group membership associated with device **160**, or a combination thereof. In other examples, the access profiles in database of access profiles **145** may be based upon other, or additional, information, qualities, or characteristics. In some examples, both network **140** and service offering manager **172** contain an access profile database **145**. In other examples, just one of network **140** or service offering manager **172** has an access profile database **145**. Access profile database **145-***b* may be queued when the wireless communication system receives a request (e.g., from device **160**) associated with a service offering, in order to determine if access to the service offering is authorized. Once device **160** selects one or more access profiles, those access profile may be associated with device **160**. When device **160** requests communication with a network destination **141**, the one or more access profiles associated with device **160** (which may now be device-specific profiles **149**) may be queued to determine if communication is permitted.

[0031] Service offering manager **172** may also include a portal builder **188** and a service offering request manager **184**. Portal builder **188** may provide a back-end ability to manage and control which services are deployed to different users or devices **160** associated with mobile platform **170** or an associated carrier. Which services are offered to each user or device **160** may be based on a set of rules unique to the user or device **160**, such as the access profiles. Portal builder **188** may generate a display for a user to browse service offerings at device **160**.

[0032] Service offering request manager **184** may process requests for service offerings by a device **160**. In some examples, service offering request manager **184** may process selections of service offerings.

[0033] The on-board communication system of mobile platform **170** may provide communication services for devices **160** via modem **178**. Devices **160** may connect to and access network **140** and service offerings through modem **178**. Devices **160** may be mobile devices such as laptops or cellular phones or may be other types of devices within mobile platform **170**. Devices **160** may communicate with the modem **178** via network access unit **164**, which may provide network services such as DNS, IP address management (e.g., dynamic host configuration protocol (DHCP)), network address translation (NAT), and the like. Devices **160** may connect with network access unit **164** via wired or wireless connections. For example, devices **160** may connect to network access unit **164** via wireless access point (WAP) **166**, which may provide wireless connectivity within mobile platform **170**. The wireless connectivity may be, for example, according to a wireless local area network (WLAN) technology such as IEEE 802.11 (Wi-Fi), or other wireless communication technology.

[0034] In the example of FIG. **1**, two devices **160**-*a* and **160**-*b* are shown as wireless user devices and two devices **160**-*c* and **160**-*d* are shown as fixed devices. A device **160**-*a* or **160**-*b* may be a mobile device such as a wireless device, a remote device, a handheld device, a subscriber device, or some other suitable terminology, where the "device" may also be referred to as a unit, a station, a terminal, or a client. Devices **160**-*a* or **160**-*b* may also be a personal electronic device such as mobile phones, personal digital assistants (PDAs), tablet computers, laptop computers, personal computers, other handheld devices, netbooks, notebook computers, display devices (e.g., TVs, computer monitors, etc.), printers, or other types of customer premises equipment (CPE), and the like. Devices **160**-*c* and **160**-*d* may be devices installed within mobile platform **170**, such as a display screen or monitor on the back of a seat. In other examples, other numbers of devices **160** may be included.

[0035] A device **160** may generate a request for a service offering. For example, a user associated with device **160** may use connectivity provided by WAP **166**, network access unit **164**, and satellite terminal **150** to obtain a service offering and browse any available content items or other services. Upon selecting a service offering to accept, the user may prompt device **160** to send a request for the service offering, which may identify any selected content items or services and any other relevant information including requested features, duration of the content, related content, recommended content, and the like. Network **140** may provide the service offering to device **160** and may also provide the service to device **160**. Network **140** may include at least one service offering **166**.

[0036] Satellite terminal **150** may provide a plurality of service offerings, such as a network access service (e.g., Internet access, etc.) or other communication services (e.g., broadcast media, etc.), that are particular to each device **160** via wireless communications system **100** (including the satellite communications system). In some examples, satellite terminal **150** provides for two-way communications between the device **160** and network(s) **140** via the communications satellite **121** and the gateway **130**. In other examples, other types of devices may be connected to the satellite terminal, such as any equipment located at a premises of a subscriber, including routers, firewalls, switches, private branch exchanges (PBXs), Voice over Internet Protocol (VoIP) gateways, and the like. In some examples, devices **160** may include an installation device (e.g., a hand-held installation device, a laptop computer, etc.) configured to perform various procedures for establishing, maintaining, and/or troubleshooting a communications link between communications satellite **121** and satellite terminal **150**.

[0037] Although examples of a satellite terminal communications antenna **152** described herein use two-way satellite communications for illustrative purposes, the techniques described herein are not so limited. For example, the hardware and techniques described herein could be used on antennas for point-to-point terrestrial links and in some examples may not be limited to two-way communication.

[0038] The wireless communication system may use techniques described herein to provide curated service offerings to each device **160** onboard a mobile transport. The service offerings may be al a carte offerings that a user associated with the device may select from. The wireless communication system may provide communications between the onboard devices and ground-based networks that hosts the services, applications, or both in the service offerings. These techniques may reduce costs, provide more personalized and relevant services or applications to passengers, improve streaming functionality, reduce size, weight, and power limitations onboard mobile platforms, provide a wider range of media content items to passengers, or reduce network traffic.

[0039] FIG. **2** shows a diagram of an in-transport media delivery system **200**, in accordance with various aspects of the present disclosure. In-transport media delivery **200** includes a wireless communication system, including at least one satellite **121**-*a*, a network **140**-*a*, and communication equipment of a mobile transport **170**-*a*. The one or more networks **140**-*a* may be outside of the wireless communication system and may be ground-based, terrestrial networks. The satellite **121**-*a*, the mobile platform **170**-*a*, and the one or more networks **140**-*a* may be an example or include aspects of the satellite **121**, the mobile platform **170**, and the one or more networks **140** of FIG. **1**, respectively.

[0040] FIG. **2** shows another view of the wireless communications system **100** including a detailed block diagram of one example embodiment of the network access unit **164**-*a*. Many other configurations of network access unit **164**-*a* are possible having more or fewer components. While shown as a single physical computing device in FIG. **2**, in some aspects, one or more of the functions discussed below with respect to the network access unit **164**-*a* may be performed by multiple different physical computing devices. In some aspects, the network access unit **164**-*a* may include fewer functional components than shown in the example of FIG. **2**. Moreover, the functionalities described herein can be distributed among the components in a different manner than described.

[0041] In some aspects, network access unit **164**-*a* provides gateway functionality between the personal electronic devices **160**-*e* on mobile transport **170**-*a* and network **140**-*a* or network destinations **141** discussed above with respect to FIG. **1**. In some examples, personal electronic devices **160**-*e* may be examples of one or more aspects of devices **160** as described in FIG. **1**. In some aspects, all network traffic transmitted by and/or received by personal electronic devices **160**-*e* may be processed by the network access unit **164**-*a*. For example, in some aspects, network traffic transmitted to a personal electronic device **160**-*e* from devices within network **140**-*a* may be electronically received by network access unit **164**-*a*, which then forwards or retransmits the network traffic to personal electronic devices **160**-*e* according to one or more policies. Similarly, in some aspects, network traffic transmitted to devices on network **140**-*a* by personal electronic devices **160**-*e* may be electronically received by network access unit **164**-*a*, which then may forward or retransmit the network traffic to destinations indicated in the network traffic (for example, to destinations within network **140**-*a*) based on one or more policies.

[0042] Consistent with FIG. **1**, an in-transport network access unit **164**-*a* is in communication, via satellite **121**-*a* (or other suitable communications network, as described above) and other components of a two-way communication system such as that shown in FIG. **2**, with a terrestrial based network **140**-*a*. Network access unit **164**-*a* may also be in communication with personal electronic devices **160**-*e* and, in some aspects, a transport management computer **202**. In FIG. **2** and the following discussion, gateway **130** and some components (e.g., satellite terminal **150**, transceiver **176**, modem **178**, and WAP **166**, etc.) of the wireless communication system **100** discussed above with respect to FIG. **1** are omitted to avoid over complication of the drawing.

[0043] The illustrated aspect of network access unit **164**-*a* includes an electronic hardware processor **180**-*a* and a network interface **210**. Processor **180**-*a* may be in communication with network interface **210** via an electronic bus within network access unit **164**-*a*. Processor **180**-*a* may communicate with network interface **210** to transmit and/or receive packets over a network, such as a network providing connectivity to the wireless access point **166** and/or modem **178** discussed above with respect to FIG. **1**. Processor **180**-*a* may also communicate over network interface **210** to exchange network messages with a personal electronic device **160**-*e* and/or a transport management computer **202**.

[0044] Network access unit **164**-*a* also includes an account provisioner **215**, a web server **216**, a traffic classifier **217**, a rules engine **220**, a traffic scheduler **220**, and a service offering manager

172-*a*. The account provisioner **215**, web server **216**, traffic classifier **217**, rules engine **220**, traffic scheduler **220**, and service offering manager **172-***a* may be portions of a volatile or stable storage, such as a virtual or physical memory space accessible to processor **180-***a*. The account provisioner **215**, web server **216**, traffic classifier **217**, rules engine **218**, traffic scheduler **220**, and service offering manager **172-***a* may include binary data defining instructions that configure the processor **180-***a* to perform various functions. For example, the account provisioner **215** may include instructions that configure processor **180-***a* to provision access to network **140-***a* by one or more of the personal electronic devices **120***a-n*.

[0045] Web server **216** may store instructions that configure processor **180-***a* to provide functionality associated with delivering electronic content via web standards, such as html/http. Traffic classifier **217** may include instructions that configure processor **180-***a* to classify network traffic received by network access unit **164-***a*. For example, traffic classifier **217** may determine whether traffic is email, web browsing traffic, video streaming traffic, messaging traffic, or other types of traffic. This information may be used to determine whether the traffic is allowed by a traffic policy as discussed in more detail below.

[0046] Rules engine **218** may include instructions that configure processor **180-***a* to execute one or more rules that assign a traffic policy to a personal electronic device **160** based on one or more characteristics of the network device. Traffic scheduler **220** may include instructions that configure processor **180-***a* to schedule network traffic generated by personal electronic devices **160-***e*. Instructions in one or more of account provisioner **215** and/or traffic schedule **220** may configure processor **180-***a* to read data from a policy database **225**. For example, in some aspects, processor **180-***a* may read data from policy database **225** and rules database **228** in order to determine a traffic policy to apply to network traffic generated by one or more of personal electronic devices **160-***e*.

[0047] While FIG. **2** shows network access unit **164-***a* as one physical device, one of skill in the art would understand that in some aspects, the functions discussed above and below relating to network access unit **164-***a* may, in some implementations, be implemented on multiple physical devices. For example, in some aspects, functionality associated with each of the account provisioner **215**, web server **216**, traffic classifier **217**, rules engine **218**, traffic scheduler **220**, and service offering manager **172-***a* may each be provided on a separate physical device having its own dedicated electronic hardware processor, memory, and network interface. Additionally, in some implementations, policy database **225**, rules database **228**, and access profile database **145-***c* may also be implemented on separate devices from one or more of the account provisioner **215**, web server **216**, traffic classifier **217**, rules engine **218**, and traffic scheduler **220**. How the functionality discussed above and below is partitioned across one or multiple physical hardware devices does not substantially affect the methods and systems disclosed herein. While FIG. **2** shows policy database **225**, rules database **228**, and access profile database **145-***c* included as part of network access unit **164-***a*, in some other implementations, these components may be accessible via network **140-***a*. In some of these aspects, network access unit **164-***a* may be configured to cache at least a portion of policy data **225**, rules database **228**, and/or access profile database **145-***c* in a local memory to provide for reliable and performant access to data contained therein.

[0048] FIG. **3** shows a process flow diagram **300** of an example for providing service offerings, in accordance with aspects of the present disclosure. The process flow diagram **300** shows example communications between a service offering manager **172-***b* (for example of a wireless communication system that includes one or more satellites), a network **140-***b*, and a device **160-***f*. The wireless communication system may include aspects of the wireless communication systems of FIGS. **1** and **2**, for example the service offering manager **172-***b*, the network **140-***b*, and the device **160-***f* may be aspects of the service offering manager **172** and **172-***a*, the network **140** and **140-***b*, and the devices **160-***a* to **160-***d* and **160-***e* of FIGS. **1** and **2**, respectively. In some examples, the device **160-***f* may be located on a mobile platform, such as an aircraft

[0049] The network **140-***b* may represent one or more networks. The network **140-***b* may provide information related to one or more service offerings **305** to the service offering manager **172-***b*. The service offerings may be an offer for an application or service that is personalized for a particular carrier, mobile transport, user, or device. In some examples, the service offering is based on a route, a sector, a base, or an area of an associated carrier. In some examples, the service offerings may be dynamically changed to personalize them to a carrier, a user, or a device. A carrier may be any person or company that transports people or goods, such as an airliner that has a fleet of aircraft, or a company with a fleet of ships, for example.

[0050] The service offerings may be associated with an access profile for a plurality of devices **160-***f*. That is, each service offering may be authorized for a particular device **160-***f* based on an access profile of the device **160-***f*. The access profile may define a set of rules for a plurality of service offerings, and may be based on one or more characteristics of the device **160-***f*. For example, the access profile may show that one or more service offerings may be authorized for the device **160-***f* based on an identity of a user associated with the device **160-***f*, a subscription of the device **160-***f*, a service provider of the device **160-***f*, one or more characteristics of a mobile platform carrying the device **160-***f*, specific to a flight or other travel route, communication information, a reservation identifier, a ticket identifier, an individual membership associated with the device **160-***f*, a group membership associated with the device **160-***f*, or a combination thereof. In other examples, the access profile may be based upon other, or additional, information, qualities, or characteristics.

[0051] Once a service offering manager **172-***b* is provisioned with or has access to the one or more service offerings to make available to passengers, the service offering manager **172-***b* may provide information related to the service offerings to the device **160-***f* in a service offering information signal **310**. Information in the service offering information signal **310** may include identifying information for each service offering, including what type of service is offered, any applications or services that are included, the terms of the service offering, any expiration of the service offerings, and the like. The service offering information signal **310** may be provided to the device **160-***f* upon request or upon connection of the device **160-***f* to a network of the mobile platform. The service offering information signal **310** may be generated at the service offering manager **172-***b* or at the network **140-***b*.

[0052] The device **160-***f* may send a request for a service offering signal **315** to the service offering manager **172-***b*. For example, a user associated with the device **160-***f* may browse the one or more service offerings available for the device **160-***f*. Upon selecting a service offering to accept, the user may prompt the device **160-***f* to send the request for the service offering signal **315**, which may identify the selected service offering and any other relevant information including requested features, duration, and the like.

[0053] Upon receiving the request for the service offering signal **315**, the service offering manager **172-***b* may identify an access profile associated with the particular selected service offering. That is, the service offering manager **172-***b* may identify an access profile from a database of access profiles at **320**, wherein the particular access profile is associated with the service offering. Based upon the relevant access profile, the service offering manager **172-***b* may determine that communications are permitted between the device **160-***f* and a respective set of network destinations associated with the identified access profile. In some examples, the service offering manager **172-***b* may approve the service offering request at **325** based on the access profile. In other examples, the service offering manager **172-***b* may determine that the access profile does not allow for approval of the service offering for the device **160-***f*. In such a case, the service offering manager **172-***b* may deny the request for the service offering signal **315**.

[0054] If the service offering manager **172-***b* determines that the service offering is approved based upon the access profile, the service offering manager **172-***b* may send an authorization signal **330** to the network **140-***b*. The network **140-***b* and the service offering manager **172-***b* may establish a communication session **335**, and in turn, the service offering manager **172**-*b* may establish a communication session **340** with the device **160**-*f*. In some examples, the communication sessions **335** and **340** may be micro-sessions. A micro-session may be a session that is restricted to the service offering, and may also be restricted to the respective set of network destinations associated with the identified access profile. The communication sessions **335** and **340** may be two-way communication sessions, where communications **345** between the device **160**-*f* and the network **140**-*b* are relayed via the service offering manager **172**-*b*. In other words, the service offering manager **172**-*b* provides a communication service to the device **160**-*f* within a mobile platform via the wireless communication system.

[0055] FIG. **4** shows an example screenshot **400** of a portal **405** for a plurality of service offerings **410**, in accordance with aspects of the present disclosure. The portal **405** may be displayed at a display screen of a device, such as a device **220** as in FIGS. **2** and **3**, when a service offering is received, such as at **310** in FIG. **3**. The portal **405** may be displayed at a mobile device onboard a mobile transport or may be displayed at a screen of a terminal fixed on the mobile transport. The screenshot **400** is merely one example of many possible different screenshots and alternatives for the portal **405**.

[0056] The portal **405** may include an identifier **435** that shows that the portal **405** is for establishing one or more micro-sessions. As used herein, a micro-session may include access to only a portion of the Internet or other network-based service, wherein the portion is related to an application or service identified in a service offering. The portal **405** may include a list of service offerings **410**, which may include prices **420**. Example service offerings are listed in the portal **405**, as well as example prices. For example, a messaging service may be listed as free, along with 8 different options for different messaging applications that a user can select. The different options may be selected among popular messaging services at the time, messaging services typically used in a region or along a route of the carrier or mobile transport, personalized user history associated with the device, or the like. Other service offerings shown in the example of the portal **405** may include social networking, a video service, a music application, a spreadsheet application, or a mapping application. In other examples, other service offerings may be included, which may have more or fewer service offerings than shown in the example of FIG. **4**.

[0057] The portal **405** may also include one or more request options **415**, such as radio buttons, where the user of the device may select a service offering. As shown in FIG. **4**, request option **415**-*a* has been selected by placing an X in radio button **415**-*a*, which selects the associated service offering named music. A submit option **440** may be selected to provide a request for the service offering to the wireless communication system. If authorized, the device may be able to connect to the service offering via the wireless communication system. If submit option **440** is selected, then the device **160**-*f* may request a service offering, such as at **315** of FIG. **3**.

[0058] While the example portal **405** is shown simply as a line drawing, other examples may include graphics, videos, colors, graphics interchange formats (GIFs), icons, or the like. Other options or features may also be included, such as options to submit, cancel, update service offerings, or the like. In some examples, the portal **405** is a webpage, rendered by a web browser. In other examples, the portal **405** is a graphical user interface (GUI) for another type of application.

[0059] In some examples, the portal **405** may be generated at a portal builder of a service offering manager, such as the portal builder **188** as shown in FIG. **1**. The portal builder may be hosted within the wireless communication system, which may be at a satellite gateway or a satellite terminal, for example. The portal builder may provide a back-end ability to manage and control which services are deployed to different users or devices associated with the carrier. Which services are offered to each user or device may be based on a set of rules unique to the user or device, such as the access profile. The access profile rules may be changed or updated based on changes at the provider offering the service or changes to something associated with the carrier, user, or device (such as, for example, a subscription status, a service provider, a location, a route, local competitors, market changes, license changes, applications installed on the device, etc.). Prices may also differ between different devices as well, which may be based on things like enrollment in a loyalty program, the identity of a service provider of the device, existing subscriptions, current promotions, and regions, among other factors.

[0060] The portal **405** may provide a back-end technique for dynamically determining personalized service offerings for a plurality of devices. The service offerings that are displayed on the portal **405** may be personalized for each device, which may, for example, be based on current subscriptions and installed applications. The portal **405** may be different for different users. The portal **405** may provide a more unique and streamlined experience for a user, such as offering services that are relevant to that user and at relevant prices. A user may save time browsing through many different service offerings when using the techniques described herein, since more relevant service offerings may be more prominently displayed. In some examples, a user may also save money by only paying for the particular service that they want to use in a micro-session, instead of having to pay a higher price for full access to the network. Traffic in the network may also be reduced using these techniques.

[0061] FIG. **5** shows a flowchart illustrating an exemplary method **500** for personalized connectivity service offerings with respect to a wireless communication system, in accordance with aspects of the present disclosure. In some examples, the wireless communication system may be or include aspects of a wireless communication system as described with respect to FIGS. **1-4**.

[0062] At **505**, one or more service offerings may be received at the wireless communication system from one or more networks. The service offerings may be valid for a specific time period or may be real-time offers. The service offerings may be offers for device-specific access to one or more applications or services provided by a respective set of network destinations within the network. The service offerings may be generated based on a carrier, a route, a region, etc. of the mobile platform. At **510**, the method **500** includes providing the service offerings to a plurality of devices on the mobile platform. In some examples, an access profile database may be consulted before the service offerings are provided to the plurality of device in order to further personalize the service offerings to the specific devices.

[0063] At **515**, the method **500** includes determining whether a request has been received from a device for a service offering. If not, the method **500** continues to receive updated offerings from the network and provide those offerings to the devices. If a request for access to a service offering is received, the method **500** proceeds to **520**.

[0064] At **520**, the access profile database is queried to determine the rules associated with the particular device accessing the particular service offering. At **525**, the method **500** determines whether the access profile associated with the service offering and the device allow connection to the service offering. If the service offering complies with the access profile, the method **500** may proceed back to updating its service offerings or providing updated service offerings to the device. However, if the service offering does comply with the access profile, the device may be authorized by the wireless communication system to use the service offering. At **530**, a communication session may be established between the device and the network, with the wireless communication system acting as a relay. The communications session may be a micro-session.

[0065] At **535**, the method **500** may include the wireless communication system relaying communications between the device and the network during the session. If the method **500** includes receiving a request to end the session at **540**, either from the device or the network, the session may be ended at **545**. In other examples, the session may be associated with a session timer, and the session may time out after expiration of the session timer. If not, the method **500** continues to support communications between the device and the network until the session is ended. The method **500** may repeat or be performed in parallel for different devices connecting to the wireless communication system, as well as for updated service offerings.

[0066] FIG. **6** shows a flowchart illustrating an exemplary method **600** for personalized connectivity service offerings with respect to a device, in accordance with aspects of the present disclosure. In some examples, the device may be a device **160** or a device **220**, such as a mobile device or any other type of device, as described with respect to FIGS. **1-3**.

[0067] At **605**, the method **600** may include establishing a connection to a wireless communication system, for example a wireless communication system including one or more aspects described with respect to FIGS. **1-5**. This may be, for example, a mobile device connected to a Wi-Fi network on a mobile transport character, for example the Wi-Fi network in communication with a satellite communication system to connect with a network. The device may receive service offerings from the wireless communication system and display (or orate, etc.) them at **610**. A user of the device may select a service offering to use, and the device sends a request for the particular service offering to the wireless communication system at **615**.

[0068] If, at **620**, authorization for the service offering is denied, the method **600** may go back to **615**, where the user may be able to select another service. However, if authorization for the service offering is accepted, the method proceeds to **625**. At **625**, the device engages in a communication associated with the service offering that is hosted by the wireless communication system, which may be a micro-session. The device may continue to engage in the micro-session until an indication that the service offering is finished, at **630**. The service offering may be finished if the user decides to end use of the service, the service offering expires, or the provider of the service offering indicates the service is over. At **635**, the communication session may be ended based upon the indication.

[0069] In one use case, for example, a user may be onboard an aircraft flying a particular route. The user may connect their mobile phone to the Wi-Fi network onboard the aircraft. Upon connecting to the Wi-Fi network, the wireless communication system also connected to the Wi-Fi network, or including the Wi-Fi network, may identify one or more service offerings and compare them to an access profile for the mobile phone. Based on the service offerings and the access profile, the wireless communication system may generate, at the back-end, a portal for the mobile phone that identifies the particular, personalized service offerings. The wireless communication system may present the service offerings in the form of the portal to the mobile phone. The user may be able to access the portal on their mobile phone, browse the service offerings, and select which service offerings they would like to use. The mobile phone requests access to the associated service offering via on-board Wi-Fi, which relays the request to the wireless communication system. The wireless communication system, specifically a satellite terminal (which may be co-located at the aircraft or on the ground), may further confirm the mobile device's access to the service offering. Once approved, the wireless communication system facilitates communications between the mobile device and the network that hosts the service offering. The mobile device gets access to the service offering, but not necessarily other services or network sites, via the relayed communications.

[0070] FIG. **7** shows a flowchart illustrating an exemplary method **700** for personalized connectivity service offerings with respect to a network hosting the service offering, in accordance with aspects of the present disclosure. In some examples, the network may be, for example, a network **140** as described with respect to FIGS. **1-3**.

[0071] At **705**, the method **700** may include providing a set of service offerings to a carrier via a wireless communication system. Each service offering may be associated with a set of respective network destinations. The service offerings may be for an online application or service. The network may provide the set of service offerings periodically, upon request, or whenever there is a change in the service offerings, for example. The set of service offerings may detail what applications and services are being offered, prices, any restrictions, etc., and may be based upon particular characteristics of the associated carrier or mobile transport.

[0072] At **710**, the network may receive an authorization message regarding a particular device that wants access to a particular service in the service offerings. This may be received immediately after providing the set of service offerings, or may be received at a later time. The authorization message may indicate that the device has authorization, via the relevant access profile, to access the particular service.

[0073] At **715**, a communication session is established between the device and the network via the wireless communication system, which may be or include a satellite communication system. The communication session may be a micro-session, which means only those network destinations associated with the particular service may be accessible by the device. The network communicates with the device via the wireless communication system at **720**, until the communication session is ended at **725**. The communication session may be ended by the device, the network, or the wireless communication system (e.g., a device of the wireless communication system).

[0074] FIG. **8** shows a flowchart illustrating an exemplary method for personalized connectivity service offerings, in accordance with aspects of the present disclosure. Some or all of the steps of the exemplary method **800** may be performed by various devices of a wireless communications system, for example one or more of a satellite terminal **150**, a communications satellite **121** of a satellite communications system, a gateway **130**, and/or a network device **141** of a satellite communications system, as described with reference to FIGS. **1-3**. In the description of method **800**, one or more devices of the wireless communications system may perform each of the described steps. In some examples, one device of the wireless communications system may perform the steps. In other examples, one or more steps may be performed by a first device, while a second device may perform one or more other steps of method **800**.

[0075] At **805**, the method **800** may include providing a communication service to a plurality of devices within a mobile platform via the wireless communication system. In some examples, the mobile platform is associated with a carrier that defines the access profiles for the mobile platform, wherein the carrier is associated with a plurality of mobile platforms. In some examples, **805** may be performed at least in part by a network manager **1020**, a service offering request manager **1015**, and/or a transmitter **1095** as described in FIG. **10**.

[0076] At **810**, the method **800** may include maintaining a database of access profiles for accessing a network via the wireless communication system, wherein each of the access profiles corresponds to a respective service offering of a plurality of service offerings selectable via the plurality of devices, and is associated with a respective set of network destinations within the network. In some examples, **810** may be performed at least in part by an access profile database **1010** as described in FIG. **10**.

[0077] At **815**, the method **800** may include receiving a request from a device of the plurality of devices for a particular service offering of the plurality of service offerings. In some examples, **815** may be performed at least in part by the service offering request manager **1015** or a receiver **1090** as described in FIG. **10**.

[0078] At **820**, the method **800** may include identifying an access profile from the database of access profiles that is associated with the particular service offering. In some examples, the access profiles define a set of rules for the plurality of service offerings, including the particular service offering, offered at the plurality of devices within the mobile platform. In other examples, each access profile defines a set of rules for the plurality of devices to access the respective service offering. In some examples, the set of rules for access is based at least in part on at least one of a subscription, an identification, communication information, a reservation identifier, a ticket identifier, an individual membership associated with the device, a group membership associated with the device, or a combination thereof. In some examples, **820** may be performed at least in part by the network manager **1020** or the access profile database **1010** as described in FIG. **10**.

[0079] At **825**, the method **800** may include permitting communications between the device and the respective set of network destinations associated with the identified access profile. In some examples, permitting communications between the device and the respective set of network destinations associated with the identified access profile further includes permitting communications on beam resources of at least one satellite beam of a satellite communication system, wherein the wireless communication system comprises the satellite communication system. In some examples, **825** may be performed at least in part by a micro-session manager **1025** or the network manager **1020** as described in FIG. **10**.

[0080] In some examples, the method **800** further includes establishing a session for the particular service offering at the device of the plurality of devices, wherein the communications are transmitted during the established session.

[0081] In some examples, the method **800** also includes configuring a session according to the access profile associated with the particular service offering and communicating information identifying the configured session of the session to the device. Some examples include configuring the plurality of service offerings based on a characteristic of the mobile platform.

[0082] In other examples, the method **800** may also include determining the plurality of service offerings based at least in part on identifying information for at least one user associated with the device. This could be done with or without the device directly communicating the identifying information. For example, a preference profile may be built based on a past use of the device or another characteristic. The preference profile may be used to determine which services to allow the device to select from.

[0083] In additional examples, the method **800** further includes communicating at least a portion of data associated with the particular service offering from a network destination of the respective set of network destinations, wherein the network destination is outside of the wireless communication system. In yet more examples, the method **800** may include updating a device-specific profile database to include an indication of the request for the particular service offering from the device.

[0084] In some examples, the method **800** further includes receiving a request from a second device of the plurality of devices for a different service offering of the plurality of service offerings. The method **800** may further include permitting communications between the second device and a respective second set of network destinations associated with a second identified access profile associated with the different service offering. In some examples, each device receives a unique service offering or set of service offerings.

[0085] In other examples, the method **800** may include maintaining a device-specific profile database, wherein each of the device-specific profiles is associated with a respective one of the plurality of devices. The method **800** may further include in response to receiving the request from the device, updating the device-specific profile that is associated with the device to include an indication of the request for the particular service offering from the device. In additional examples, the method **800** may also include permitting the communications if the indication of the request for the particular service offering is within the device-specific profile that is associated with the device.

[0086] FIG. **9** shows a block diagram illustrating a satellite communications environment **900**, in accordance with aspects of the present disclosure. The satellite communications environment **900** may be an example of the satellite communications environment **100** or **200** described with reference to FIG. **1** or **2**, respectively. The satellite communications environment **900** includes a multi-user access terminal **950**, satellite **121**-*c* (for example as part of a satellite communication system, and device **160**-*g*. In some examples, the multi-user access terminal **950** may be an example of a satellite terminal **150** described with reference to FIGS. **1**, or modem **270**. In some examples, the multi-user access terminal **950** may be a single device, for example within a mobile platform such as an aircraft **230**. In other examples, the features performed by the multi-user access terminal **950** may be divided or split between two or more devices.

[0087] The multi-user access terminal **950** may include a processor **905** and a memory **910**. The memory **910** may store computer-readable, computer-executable software or firmware code **915** including instructions that, when executed by the processor, cause the multi-user access terminal **950** to perform various functions described herein. In some examples, the code **915** may not be directly executable by the processor but may cause a computer (e.g., when compiled and executed) to perform functions described herein. The processor **905** may include an intelligent hardware device (e.g., a central processing unit (CPU), a microcontroller, an application-specific integrated circuit (ASIC), etc.). Each of the components of the multi-user access terminal **950** may communicate, directly or indirectly, with one another (e.g., via one or more buses **960**).

[0088] The multi-user access terminal **950** may be configured to communicate with one or more communications satellites (e.g., communications satellite **121**-*c*), which may be an example of aspects of a communications satellite **121** of a satellite communications system as described with reference to FIGS. **1-3**. The multi-user access terminal **950** may be configured to establish a communications link with the communications satellite **121**-*c* employing a satellite terminal communications antenna **954** and a communications signal transceiver **952**. The communications link may support bi-directional communications via forward link signals **172**-*a* and/or return link signals **173**-*a* between the multi-user access terminal **950** and the communications satellite **121**-*c*.

[0089] The communications signal transceiver **952** may include various circuits and/or processors to support receiving, transmitting, converting, coding, and/or decoding of signals **155**-*a*. For example, the communications signal transceiver **952** may include a modem to modulate the packets and provide the modulated packets to the satellite terminal communications antenna **954** for transmission, and to demodulate packets received from the satellite terminal communications antenna **954**. As illustrated in the present example, the multi-user access terminal **950** includes a single satellite terminal communications antenna **954**. However, in some cases the multi-user access terminal **950** may have more than one satellite terminal communications antenna **954**, which may be capable of concurrently transmitting or receiving multiple wireless transmissions and/or be configured to support various beamforming techniques.

[0090] The multi-user access terminal **950** may include a service offering manager **172**-*c*. The service offering manager **172**-*c* may include various circuits and/or processors to support personalized service offerings. In some examples the service offering manager **172**-*c* may include circuits and/or processors configured to receive a plurality of service offerings and apply the service offerings to a plurality of devices using access profiles.

[0091] The multi-user access terminal **950** may be configured to support communications with one or more devices **160**-*g* via signals transmitted over wired and/or wireless connection(s) **161**-*c*. The multi-user access terminal **950** may employ a local communications interface **920** supporting any number of wired and/or wireless links between the multi-user access terminal **950** and the one or more devices **160**-*g*, which may be managed by a service offering manager **172**-*c*. As illustrated by the present example, the service offering manager **172**-*c* may implemented as a separate module of the multi-user access terminal **950**, which may be configured as a standalone set of instructions (e.g., a software module having a set of instructions stored in memory, which may be a standalone portion of memory) and/or a separate processing element (e.g., a standalone CPU, microcontroller, ASIC, field-programmable gate array (FPGA), or other like integrated circuit (IC)). In other examples, some or all of the operations of the service offering manager **172**-*c* may be caused by instructions stored in the memory **910** (e.g., a portion of the code **915**), which in some examples may be performed by the processor **905**. The service offering manager **172**-*c* may control and/or configure various components of the satellite terminal perform the one or more operations of the exemplary methods **500**, **600**, **700**, or **800** described with reference to FIGS. **5-8**.

[0092] The multi-user access terminal **950** may include a satellite communications manager **930**, configured to manage various aspects of communications between the multi-user access terminal **950** and the communications satellite **121**-*c*. As illustrated by the present example, the satellite communications manager **930** may implemented as a separate module of the multi-user access terminal **950**, which may be configured as a standalone set of instructions (e.g., a software module having a set of instructions stored in a standalone portion of memory) and/or a separate processing element (e.g., a standalone CPU, microcontroller, ASIC, FPGA, or like IC). In other examples, some or all of the operations of the satellite communications manager **930** may be caused by instructions stored in the memory **910** (e.g., a portion of the code **915**), which in some examples may include steps performed by the processor **905**.

[0093] In various examples, the components of the multi-user access terminal **950** may be divided into subassemblies, where various components may be included in a subassembly either in part, or in its entirety.

[0094] FIG. **10** shows a block diagram **1000** of a service offering manager **172-***d*, in accordance with aspects of the present disclosure. The service offering manager **172-***d* may be a portion of any of a satellite terminal **150**, a device **220**, a communications satellite **121**, a gateway **130**, or a network device **141** as described with reference to FIGS. **1-3** and **9**. For example, the service offering manager **172-***d* may be a portion of a satellite terminal **150**, operating with a shared processor and memory of the satellite terminal **150**. In another example the service offering manager **172-***d* may be a standalone component of a satellite terminal **150**, receiving inputs from and sending outputs to other components of the satellite terminal **150**. In other examples, the service offering manager **172-***d* may be or form a portion of a device **160**, a communications satellite **121**, a gateway **130**, or a network device **141** which manages access rules for service offerings of one or more networks or content providers to a satellite communications system. The service offering manager **172-***d* may also be or include a processor. Each of the components of the service offering manager **172-***d* may be in communication with each other to provide the functions described herein. The service offering manager **172-***d* may be configured to receive signals from a receiver **1090**, and deliver signals to a transmitter **1095** using various techniques, including wired or wireless communications, control interfaces, user interfaces, or the like.

[0095] The service offering manager **172-***d* may include an access profile database **1005** that stores a plurality of access profiles. The service offering manager **172-***d* may also include a service offering request manager **1015**, which may perform one or more of the aspects of managing requests for particular service offerings from devices as described with reference to FIGS. **1-9**. For example, the service offering request manager **1015** may receive, as inputs from the receiver **1090**, a request for a service offering via the satellite communication system. In various examples, the service offering request manager **1015** may query the access profile database **1010** to determine whether the requested service offering is available to the requesting device.

[0096] The service offering manager **172-***d* may include a network manager **1020**, which may perform one or more of the aspects of managing a micro-session between a device and a network that provides access to the particular service offering, as described with reference to FIGS. **1-9**. The network manager **1020** may establish a session between the device and the network, and relay communications between them. The service offering manager **172-***d* may include a micro-session manager **1025**, which may perform one or more of the aspects of providing personalized service offerings, as described with reference to FIGS. **1-9**. In some examples, the micro-session manager **1025** may facilitate a micro-session between the network and the device.

[0097] Service offering manager **172-***d* may include a portal builder **1030**, which may provide a portal to the device for requesting and accessing the one or more service offerings as described with reference to FIGS. **1-9**. For example, the portal builder **1030** may determine which service offerings are applicable to a particular device, perhaps utilizing the access profile database **1010**, and present a portal to the device. In some examples, the portal may be a GUI. Service offering manager **172-***d* may include a device-specific profile database **1035**, which may provide device-specific rules for devices to access service offerings as described with reference to FIGS. **1-9**.

[0098] The components of the service offering manager **172-***d*, individually or collectively, may be implemented with at least one ASIC adapted to perform some or all of the applicable features in hardware. Alternatively, the features may be performed by one or more other processing units (or cores), on at least one IC. In other examples, other types of integrated circuits may be used (e.g., Structured/Platform ASICs, a FPGA, or another semi-custom IC), which may be programmed in any manner known in the art. The features may also be implemented, in whole or in part, with instructions embodied in a memory, formatted to be executed by one or more general or application-specific processors.

[0099] The detailed description set forth above in connection with the appended drawings describes examples and does not represent the only examples that may be implemented or that are within the scope of the claims. The term “example,” when used in this description, mean “serving as an example, instance, or illustration,” and not “preferred” or “advantageous over other examples.” The detailed description includes specific details for the purpose of providing an understanding of the described techniques. These techniques, however, may be practiced without these specific details. In some instances, well-known structures and apparatuses are shown in block diagram form in order to avoid obscuring the concepts of the described examples.

[0100] Information and signals may be represented using any of a variety of different technologies and techniques. For example, data, instructions, commands, information, signals, bits, symbols, and chips that may be referenced throughout the above description may be represented by voltages, currents, electromagnetic waves, magnetic fields or particles, optical fields or particles, or any combination thereof.

[0101] The various illustrative blocks and components described in connection with the disclosure herein may be implemented or performed with a general-purpose processor, a digital signal processor (DSP), an ASIC, an FPGA or other programmable logic device, discrete gate or transistor logic, discrete hardware components, or any combination thereof designed to perform the functions described herein. A general-purpose processor may be a microprocessor, but in the alternative, the processor may be any conventional processor, controller, microcontroller, or state machine. A processor may also be implemented as a combination of computing devices, e.g., a combination of a DSP and a microprocessor, multiple microprocessors, microprocessors in conjunction with a DSP core, or any other such configuration.

[0102] The functions described herein may be implemented in hardware, software executed by a processor, firmware, or any combination thereof. If implemented in software executed by a processor, the functions may be stored on or transmitted over as instructions or code on a computer-readable medium. Other examples and implementations are within the scope of the disclosure and appended claims. For example, due to the nature of software, functions described above can be implemented using software executed by a processor, hardware, firmware, hardwiring, or combinations of any of these. Features implementing functions may also be physically located at various positions, including being distributed such that portions of functions are implemented at different physical positions. As used herein, including in the claims, the term “and/or,” when used in a list of two or more items, means that any one of the listed items can be employed by itself, or any combination of two or more of the listed items can be employed. For example, if a composition is described as containing components A, B, and/or C, the composition can contain A alone; B alone; C alone; A and B in combination; A and C in combination; B and C in combination; or A, B, and C in combination. Also, as used herein, including in the claims, “or” as used in a list of items (for example, a list of items prefaced by a phrase such as “at least one of” or “one or more of”) indicates a disjunctive list such that, for example, a list of “at least one of A, B, or C” means A or B or C or AB or AC or BC or ABC (i.e., A and B and C).

[0103] Computer-readable media includes both computer storage media and communication media including any medium that facilitates transfer of a computer program from one place to another. A storage medium may be any available medium that can be accessed by a general purpose or special purpose computer. By way of example, and not limitation, computer-readable media can comprise RAM, ROM, EEPROM, flash memory, CD-ROM or other optical disk storage, magnetic disk storage or other magnetic storage devices, or any other medium that can be used to carry or store desired program code means in the form of instructions or data structures and that can be accessed by a general-purpose or special-purpose computer, or a general-purpose or special-purpose processor. Also, any connection is properly termed a computer-readable medium. For example, if

the software is transmitted from a website, server, or other remote source using a coaxial cable, fiber optic cable, twisted pair, digital subscriber line (DSL), or wireless technologies such as infrared, radio, and microwave, then the coaxial cable, fiber optic cable, twisted pair, DSL, or wireless technologies such as infrared, radio, and microwave are included in the definition of medium. Disk and disc, as used herein, include compact disc (CD), laser disc, optical disc, digital versatile disc (DVD), floppy disk and Blu-ray disc where disks usually reproduce data magnetically, while discs reproduce data optically with lasers. Combinations of the above are also included within the scope of computer-readable media.
[0104] As used herein, the phrase “based on” shall not be construed as a reference to a closed set of conditions. For example, an exemplary step that is described as “based on condition A” may be based on both a condition A and a condition B without departing from the scope of the present disclosure. In other words, as used herein, the phrase “based on” shall be construed in the same manner as the phrase “based at least in part on.”
[0105] The previous description of the disclosure is provided to enable a person skilled in the art to make or use the disclosure. Various modifications to the disclosure will be readily apparent to those skilled in the art, and the generic principles defined herein may be applied to other variations without departing from the scope of the disclosure. Thus, the disclosure is not to be limited to the examples and designs described herein but is to be accorded the broadest scope consistent with the principles and novel features disclosed herein.

Claims

1. A method for use in a wireless communication system, comprising: providing a communication service to a plurality of devices within a mobile platform via the wireless communication system; providing, based at least in part on one or more rules indicated in a device-specific profile that is associated with a device of the plurality of devices, a plurality of service offerings, wherein the device-specific profile is included in a device-specific profile database, and wherein each device-specific profile of the device-specific profile database is associated with a respective one of the plurality of devices; receiving a request from the device for a particular service offering of the plurality of service offerings; in response to receiving the request from the device, updating the device-specific profile that is associated with the device to include an indication of the request for the particular service offering from the device; and permitting communications between the device and a set of network destinations associated with the particular service offering, based at least in part on the device-specific profile that is associated with the device and the one or more rules, wherein the one or more rules indicate that the device is approved for the particular service offering.
2. The method of claim 1, further comprising: updating the one or more rules based at least in part on configuration information received from a management portal associated with the wireless communication system.
3. The method of claim 1, further comprising: updating the one or more rules based at least in part on a subscription status, one or more applications installed on the device, a location of the device, a service provider associated with the device, information from a provider associated with the plurality of service offerings, or any combination thereof.
4. The method of claim 1, further comprising: determining the plurality of service offerings based at least in part on identifying information for at least one user associated with the device.
5. The method of claim 1, further comprising: establishing a session for the particular service offering at the device of the plurality of devices, wherein the communications are transmitted during the established session.
6. The method of claim 1, further comprising: prohibiting, based at least in part on the one or more rules, second communications between the device and a second set of network destinations that are

different than the set of network destinations.
7. The method of claim 1, further comprising: communicating at least a portion of data associated with the particular service offering from a network destination of the set of network destinations, wherein the network destination is outside of the wireless communication system.
8. The method of claim 1, further comprising: terminating the communications based at least in part on a termination request received from the device, an expiration of the particular service offering, or a termination indication received from a service provider associated with the particular service offering.
9. A wireless communication system, comprising: a satellite providing a communication service to a plurality of devices within a mobile platform via one or more satellite beams; a device-specific profile database that comprises a plurality of respective device-specific profiles for providing, to the plurality of devices and based at least in part on one or more respective rules comprised in respective device-specific profiles of the plurality of respective device-specific profiles, a plurality of service offerings, wherein each device-specific profile of the device-specific profile database is associated with a respective one of the plurality of devices; a processor; memory in electronic communication with the processor; and instructions stored in the memory and executable by the processor to cause the wireless communication system to: receive, from a device of the plurality of devices, a request for a particular service offering of the plurality of service offerings; in response to receiving the request from the device, update a device-specific profile that is associated with the device to include an indication of the request for the particular service offering from the device; and permitting communications between the device and a set of network destinations associated with the particular service offering, based at least in part on the device-specific profile that is associated with the device and one or more rules indicated in the device-specific profile, wherein the one or more rules indicate that the device is approved for the particular service offering.
10. The wireless communication system of claim 9, wherein the instructions are further executable by the processor to cause the wireless communication system to: update the one or more rules based at least in part on configuration information received from a management portal associated with the wireless communication system.
11. The wireless communication system of claim 9, wherein the instructions are further executable by the processor to cause the wireless communication system to: update the one or more rules based at least in part on a subscription status, one or more applications installed on the device, a location of the device, a service provider associated with the device, information from a provider associated with the plurality of service offerings, or any combination thereof.
12. The wireless communication system of claim 9, wherein the instructions are further executable by the processor to cause the wireless communication system to: determine the plurality of service offerings based at least in part on identifying information for at least one user associated with the device.
13. The wireless communication system of claim 9, wherein the instructions are further executable by the processor to cause the wireless communication system to: establish a session for the particular service offering at the device of the plurality of devices, wherein the communications are transmitted during the established session.
14. The wireless communication system of claim 9, wherein the instructions are further executable by the processor to cause the wireless communication system to: prohibit, based at least in part on the one or more rules, second communications between the device and a second set of network destinations that are different than the set of network destinations.
15. The wireless communication system of claim 9, wherein the instructions are further executable by the processor to cause the wireless communication system to: communicate at least a portion of data associated with the particular service offering from a network destination of the set of network destinations, wherein the network destination is outside of the wireless communication system.
16. The wireless communication system of claim 9, wherein the instructions are further executable

by the processor to cause the wireless communication system to: terminate the communications based at least in part on a termination request received from the device, an expiration of the particular service offering, or a termination indication received from a service provider associated with the particular service offering.